

PREDICTING S&P500 REACTIONS USING FEDERAL RESERVES SPEECHES

**DATA MANAGEMENT
PROJECT**

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RESEARCH QUESTION & ECONOMIC INTUITION

Do Federal Reserve speeches influence short-term S&P 500 market reactions?

- Speeches convey information (public)
- markets react to expectations (future monetary policy), not just tone

DATA OVERVIEW

Dataset	Federal Reserve Governors Speeches (1996-2020)	Historical Data S&P 500 (1995-today)
Sources	Kaggle	Yahoo Finance
Collect	Dowload .csv file	Import via API yfinance
(Relevant) Variable	Speaker, Text, Date	Closing price, Date

FINAL RESULT

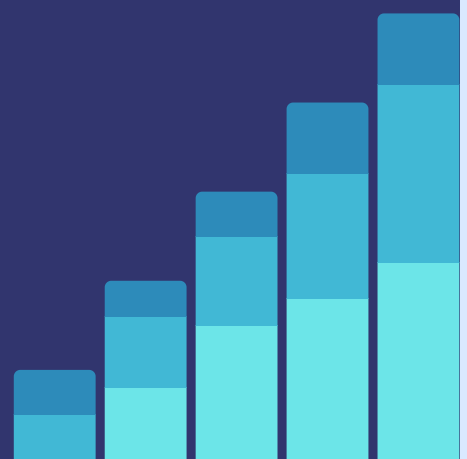
	date	title	speaker	location	text	clean_text	direction_next_day	direction_5d_ahead	direction_3d_ahead
0	1996-12-19	Supervision of bank risk-taking	Vice Chair Alice M. Rivlin	D.C.	I discovered when I joined the Board of Govern...	discovered joined board governor federal reser...	1.0	1.0	1.0
1	1996-12-06	Social security	Chairman Alan Greenspan	Pennsylvania	I am privileged to accept the Union League of...	privileged accept union league philadelphia ab...	1.0	0.0	1.0
2	1996-12-05	The challenge of central banking in a democrat...	Chairman Alan Greenspan	D.C.	The Challenge of Central Banking in a Democrat...	challenge central banking democratic society g...	0.0	0.0	1.0
3	1996-12-03	Clearinghouses and risk management	Governor Edward W. Kelley, Jr.	D.C.	It is a pleasure to be with you this morning ...	pleasure morning discus private sector payment...	0.0	0.0	0.0
4	1996-11-25	Supervisory and regulatory responses to financ...	Governor Susan M. Phillips	D.C.	Supervisory and Regulatory Responses to Financ...	supervisory regulatory response financial inno...	0.0	0.0	0.0

DESCRIPTIVE ANALYSIS

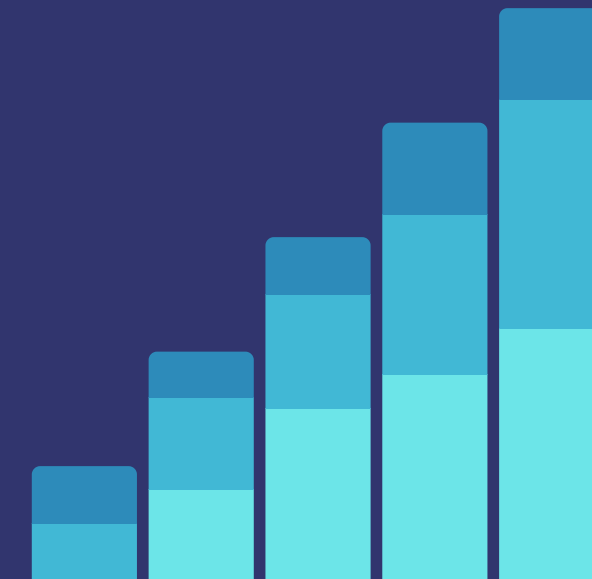
	word_count	unique_word_count	avg_word_length
count	1385.000000	1385.000000	1385.000000
mean	1882.868592	777.065704	6.848877
std	906.057261	268.832088	0.281230
min	4.000000	4.000000	5.950980
25%	1297.000000	636.000000	6.672316
50%	1858.000000	801.000000	6.835286
75%	2370.000000	935.000000	7.003347
max	6372.000000	1812.000000	9.500000

1. DESCRIPTIVE STATISTICS

- Numeric text features from clean text (unique_word, word_count and avg_word_length)

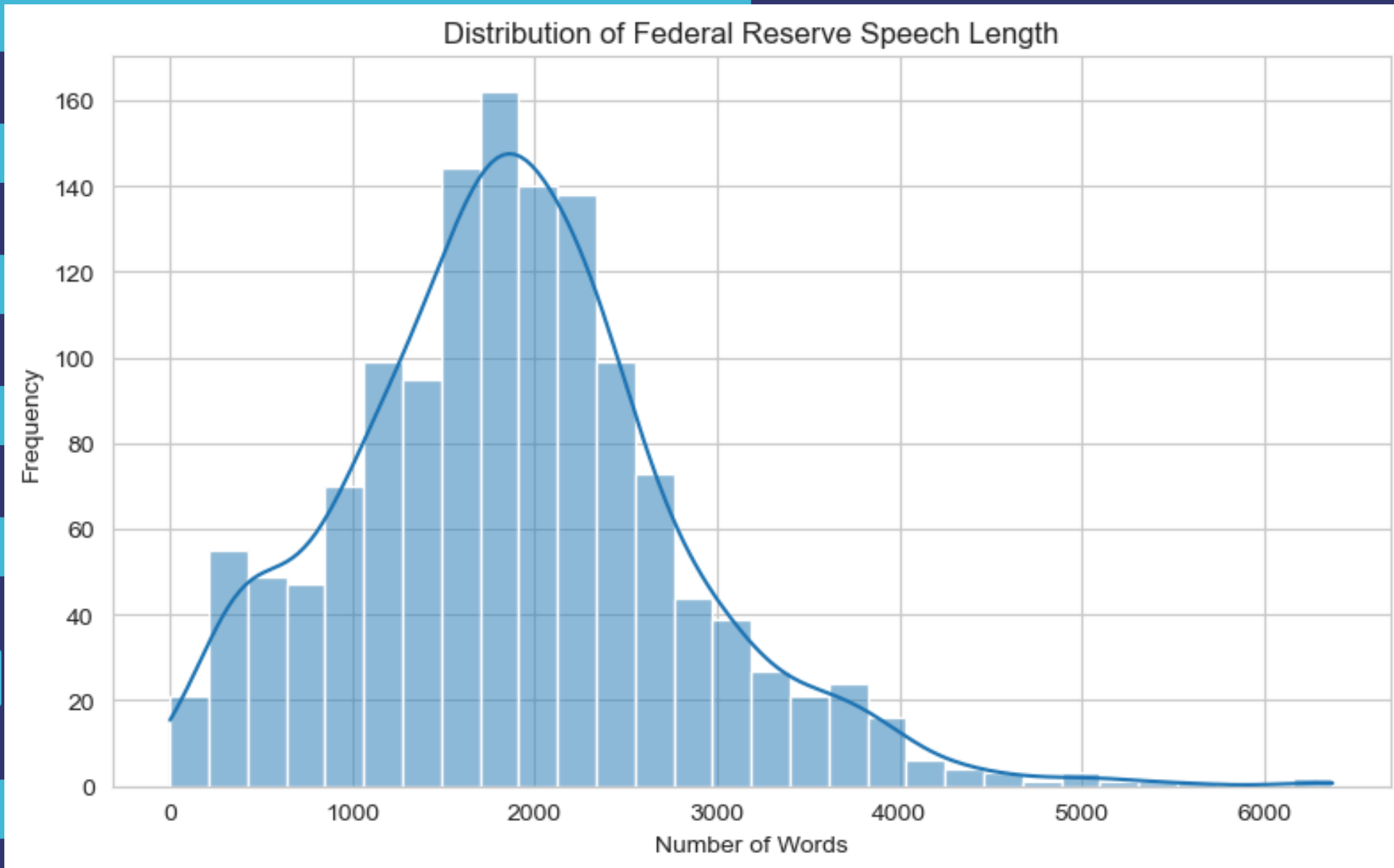


	Variable	Statistic	Market Down (0)	Market Up (1)
0	word_count	N	622.00	763.00
1	word_count	Mean	1889.85	1877.18
2	word_count	Median	1849.00	1861.00
3	word_count	Std. Dev.	928.24	888.13
4	unique_word_count	N	622.00	763.00
5	unique_word_count	Mean	774.38	779.26
6	unique_word_count	Median	797.00	806.00
7	unique_word_count	Std. Dev.	270.08	267.97
8	avg_word_length	N	622.00	763.00
9	avg_word_length	Mean	6.85	6.85
10	avg_word_length	Median	6.83	6.84
11	avg_word_length	Std. Dev.	0.28	0.28

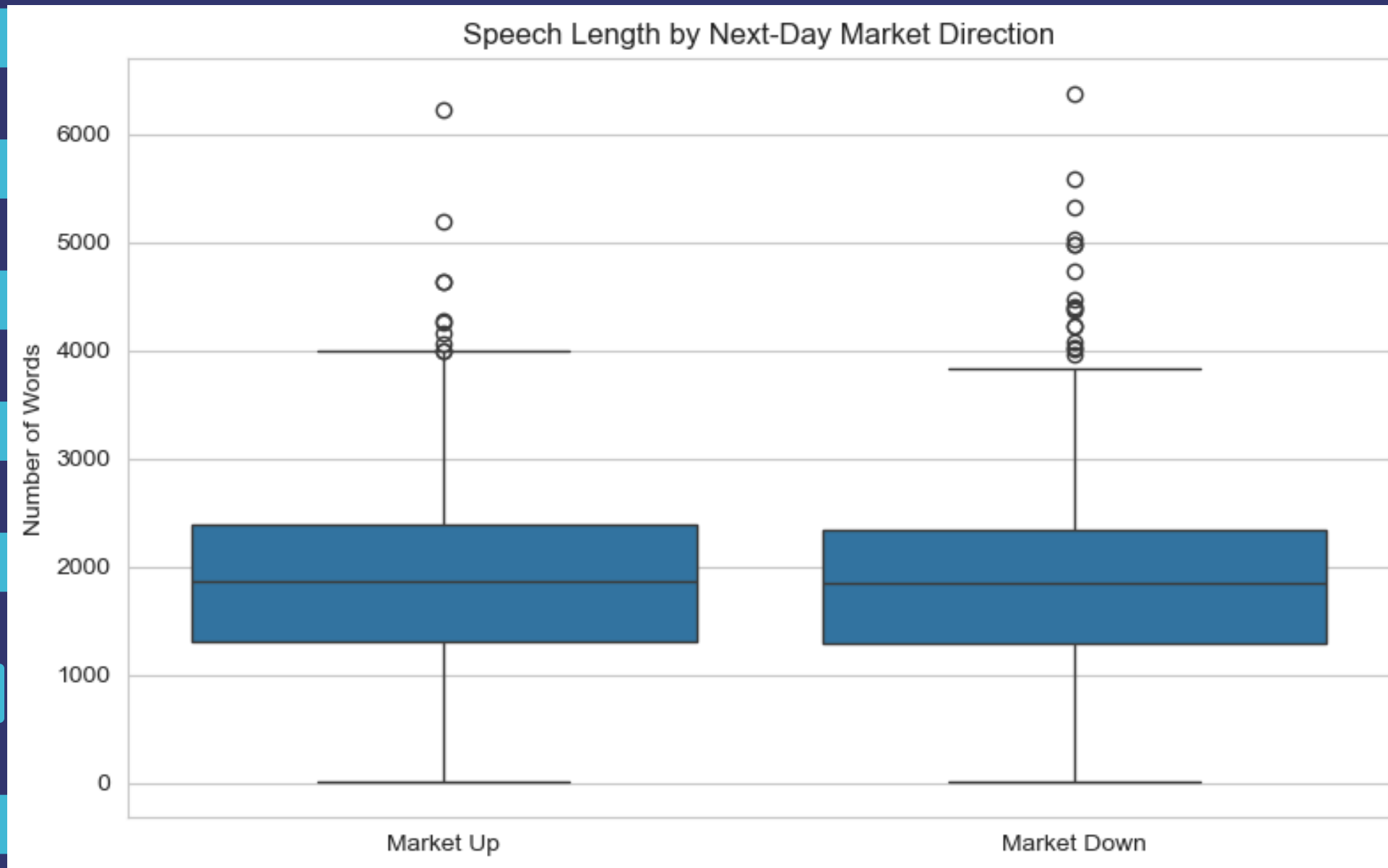


- Conditional descriptive statistetic table (Up vs Down) for the traget variables

2. GRAPHICAL ILLUSTRATION



- Distribution of speech length.

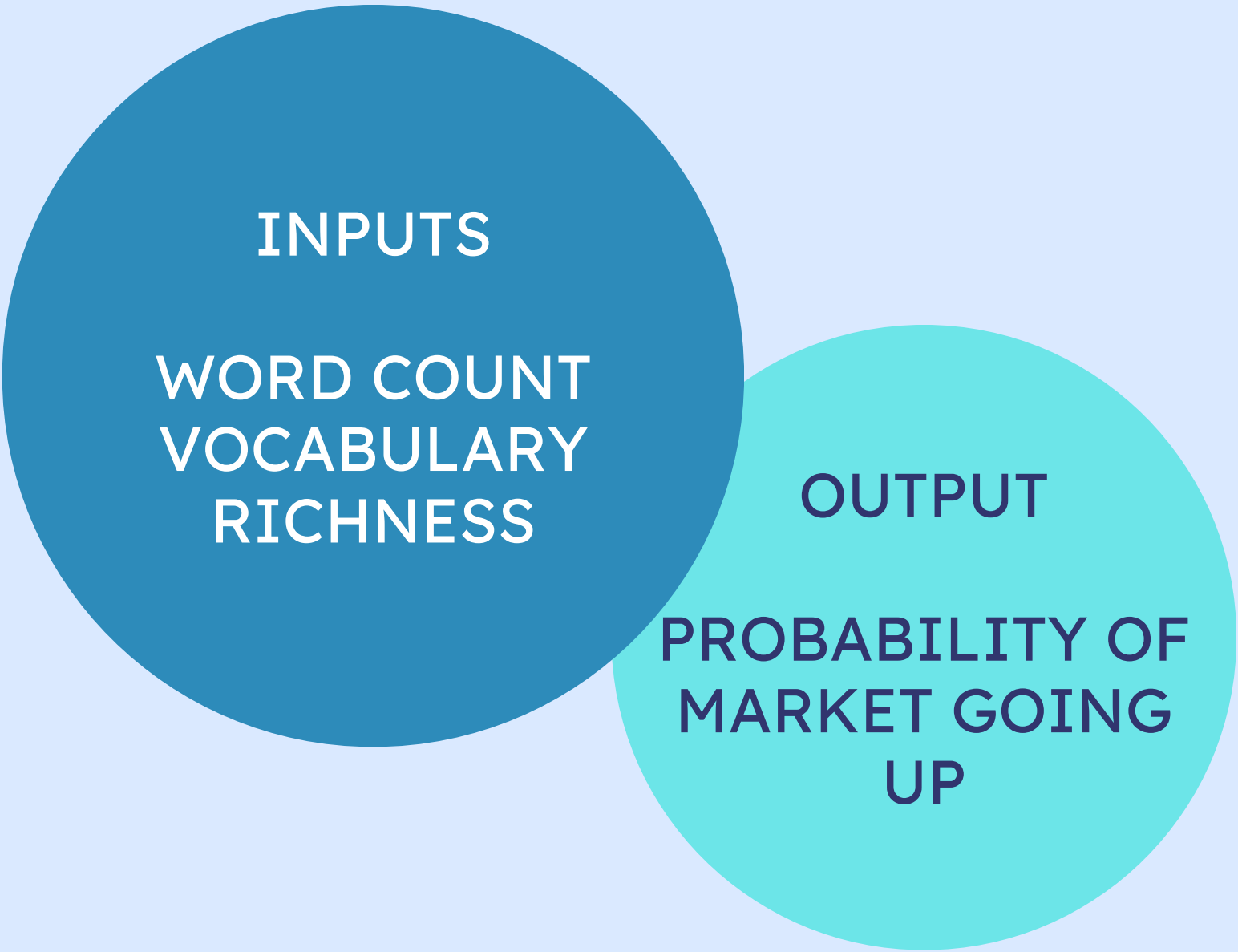


- Box Plot of text length by market direction (next day)

Descriptive statistics conclusion :
→ Need more advanced ML NLP analysis

MODELING PART

1- LOGISTIC REGRESSION



	feature	coefficient	odds_ratio
0	word_count	-0,345757	0.707684
1	unique_word_count	0.284568	1.329187
Score		ROC curve (AUC): ~48%	Accuracy: ~55%

2- LOUGHRAN McDONALD

LM financial sentiment lexicon
→ NLP → positive / negative
tone

Compute sentiment and apply
to our dataset
 $\text{Tone} = (\text{pos} - \text{neg}) / \text{total}$

Use LM tone as NLP predictor

	feature	coefficient	odds_ratio
2	lm_neg_count	0.003039	1.003044
1	lm_pos_count	-0.056516	0.945051
0	lm_tone	-0.030054	0.970393

Score	ROC curve (AUC): ~45%	Accuracy: ~58.5%
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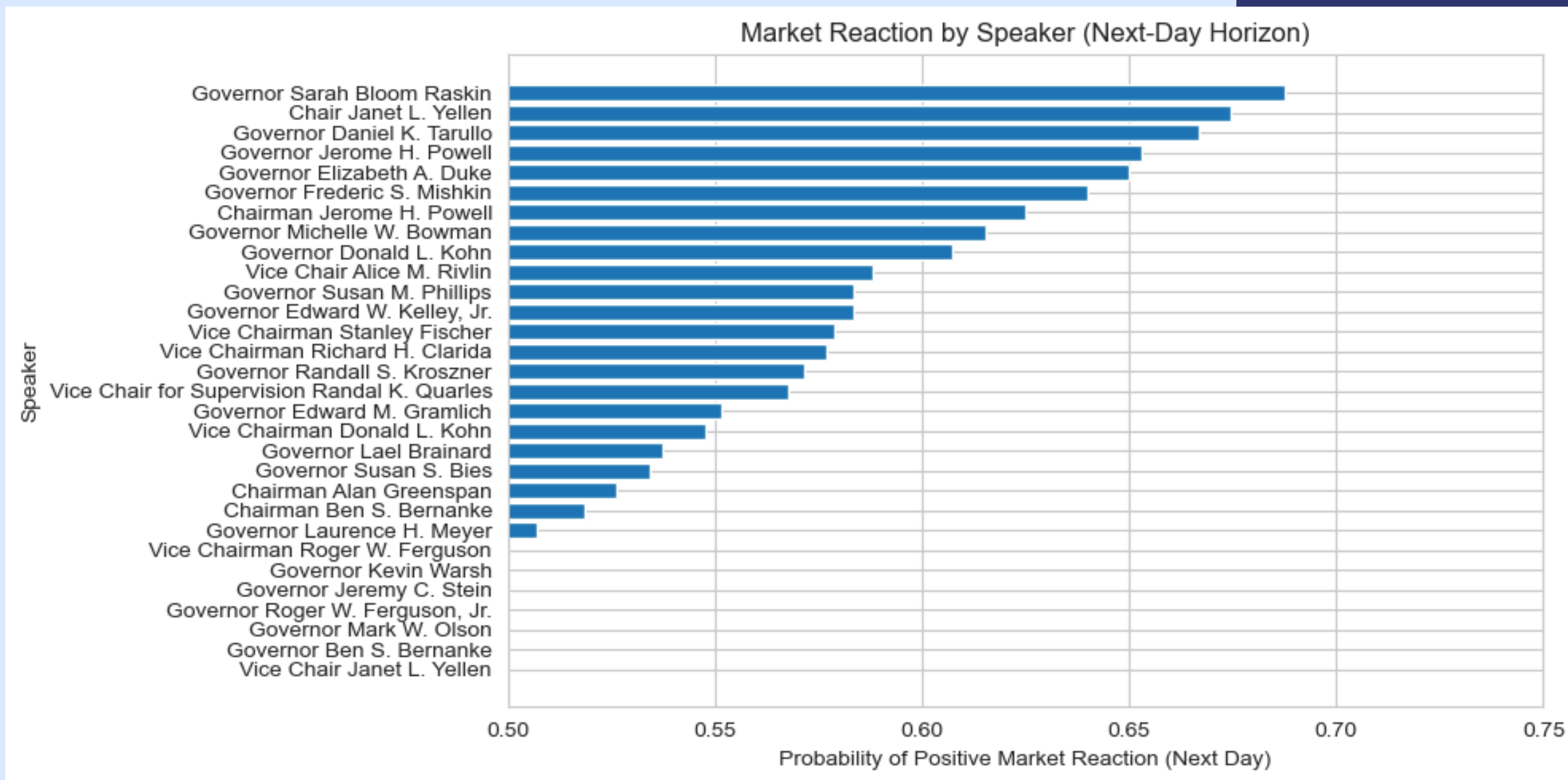
NLP RESULTS & ROBUSTNESS CHECK

- Accuracy increases slightly at longer horizons
- AUC remains modest
- Interpretation: markets may incorporate information with delay, but signal remains weak

Model	Horizon	Accuracy	AUC
Logistic	t+1	0.56	0.48
LM NLP	T+1	0.58	0.45
LM NLP	T+3	0.64	0.46
LM NLP	T+5	0.67	0.53

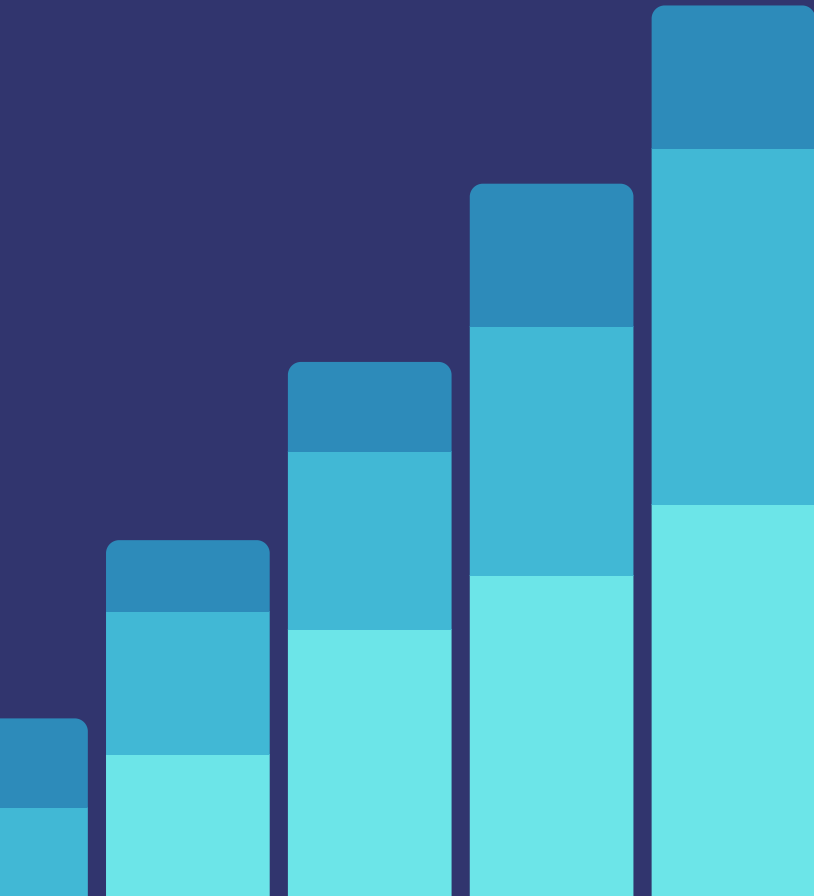
SPEAKER HETEROGENEITY

Are some speaker more
influencial than others?



- Some speakers are associated with stronger reactions
- But correlation $\neq$ causation
- Speaker identity may proxy institutional role or timing

CONCLUSION & ECONOMIC INTUITION



Can we predict S&P 500 with FED speeches?

Descriptive analysis → heterogeneity across speakers / horizon

Modeling part → limited predictive power out of sample.

RESULTS ARE
CONSISTANT

FINANCIAL
MARKETS RAPIDLY
INCORPORATE
PUBLIC
INFORMATION

HARD TO PREDICT

LIMITATION

Market direction is
influenced with many others
factors

Noisy textual signals





Loughran-McDonald Master Dictionary w/ Sentiment Word Lists

The dictionary/sentiment lists are free for use in academic research. For commercial licenses, please contact us at...

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Federal Reserve Governors Speeches 1996 - 2020

Text and metadata for public speeches given by Fed governors between 96'-20'

 [kaggle.com](https://www.kaggle.com)



S&P 500 (^GSPC) Historical Data - Yahoo Finance

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REFERENCE

THANK YOU

Q&A

DI PILLO Loucas
DEWEYS Lucas